

Not an assumption; we implicitly think that the size of the statistic is likely to be around zero when the null is true.

- p -value: the sum of the probability of equi-likely or less likely events under the null hypothesis.

the set of ^{all} possible events = $\mathbb{R}$ (i.e., T can fall into any $t \in \mathbb{R}$).

p -value

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*Not to be confused with the **P-factor**.*

In **null-hypothesis significance testing**, the **p -value**^[note 1] is the probability of obtaining test results at least as extreme as the **result actually observed**, under the assumption that the **null hypothesis** is correct.^{[2][3]} A very small p -value means that such an extreme observed **outcome** would be very unlikely *under the null hypothesis*. Even though reporting p -values of statistical tests is common practice in **academic publications** of many quantitative fields, misinterpretation and **misuse of p -values** is widespread and has been a major topic in **mathematics** and **metascience**.^{[4][5]}

$$\text{But, } \Pr(T = t) = \Pr(T = t^{\text{obs}}) = \left(\frac{10}{6}\right)^{-1}$$

for all t 's that are found.

Then, $p\text{-value} = 1$.

We have to look at $|T|$ and of the likelihoods $\Pr(|T| > t^{\text{obs}})$.
Bottom Line: Aren't we implicitly having an assumption here? $\rightarrow$ this is the basis of equi-likely & less likely

"Repeated Sampling Thought Experiment"

no No distributional assumptions.

So it's not that we're making assumptions here.